DD2434 Final Project

Variational methods for the Dirichlet process

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Abstract

The paper proposes an approximation method based on variational inference that can be applied to non-parametric settings, providing fast and deterministic solutions to the Dirichlet Process mixture models. The report first introduces the problem settings of DP mixture models and shows necessary derivations in the specific case of Gaussian. Finally, the method is tested on a toy example, simulated data, and robot data for evaluation and we thoroughly discuss its pros and cons in the discussion section.

Introduction

This article revolves around the use of Variational Inference for Dirichlet Process mixture models. In the context of probabilities, `inference` means to compute the probability distribution of one set of the variable given observations of other variables. Consider an observed variable X and a set of latent variables Z. In the Bayesian model, the likelihood $p(x|z)$ links the latent to the observations. In the case of inference, one aims at determining the posterior $p(z|x)$. The key idea behind the inference problem is to see it as an optimization problem. In variational inference, $p(z|x)$ is approximated by a distribution $q(Z)$ which belongs to a restricted set of distributions (in most cases the exponential family). The optimization consists in finding the distributions and variational parameters of $q(z)$ so that the Kullback-Leibler divergence $D_{KL}(q||p(\cdot|x))$ is minimized. Eventually, the fitted variational density serves as a proxy for the exact conditional density. In the paper, *mean-field* VI is used, we hypothesize that the distribution $q(Z)$ can be factorized as the product of subsets of latent variables.
Usually, Monte Carlo methods are used to sample DP mixture models but their being slow is a flaw variational inference can overcome. On the other hand, MC Markov Chains are asymptotically exact (the result, in the long run, tends to be the theoretical result) so VI cannot be used everywhere.

A Dirichlet distribution is a process whose realizations are probability distributions (instead of being numbers *e.g.*). The construction - called the Pòlya urn scheme - is given as the following :
Let $\eta \sim G_0$ with G_0 a non-atomic probability distribution (the base distribution) and $\alpha \geq 0$ the scaling parameter :

$$\forall n \geq 1, \eta_n = \begin{cases} \eta_i^* & \text{with probability } \frac{n_i}{n-1+\alpha} \\ \eta & \text{with probability } \frac{\alpha}{n-1+\alpha} \end{cases}$$

where :
$\eta^* := \{\eta_i,\ i < n\}$ (set of unique values taken by $\eta_{1:n-1}$), η_i^* being the ith element.
$n_i := \left|\{j : \eta_j = \eta_i^*,\ j < n\}\right|$

The DP mixture model uses the Dirichlet Process as a nonparametric prior :

$$\begin{aligned} G|\{G_0, \alpha\} &\sim DP(G_0, \alpha) \\ \eta|G &\sim G \\ X_n &\sim p(\cdot|\eta_n) \end{aligned}$$

To get a DP mixture model truncated to finite dimensional distributions, G must be defined by a *stick breaking construction* (see Sethuraman [1]):

Let $V_i \sim Beta(1, \alpha)$ and $\eta_i^* \sim G,$ for $i \in \{1, 2, \dots\}$,

$$\theta_i(v) = v_i \prod_{j=1}^{i-1}(1 - v_j)$$

$$G = \sum_{i=1}^{\infty} \theta_i(v)\delta_{\eta,\eta_i^*}$$

This definition of G makes it discrete. Let Z_n the auxiliary variable denoting the mixture associated with X_n, the data generation is to be sampled from :
1) Draw $V_i \sim Beta(1, \alpha)$ truncated to K with $V_{K-1} = 1$ and $\forall k \geq K, V_k = 0$
2) Draw $\eta_i^* \sim G$
3) For the point n:

- Draw $Z_n \sim Mult(\theta(v))$ where $\theta(v)$ is the vector of $\theta_i(v)$
- Draw $X_n \sim p(\cdot|\eta_{z_n})$

In this report, we present our derivations with respect to this model and compare the results of our VI implementation to the ones in the original paper.

Methods

Variational Inference

First, we show the necessary derivations in the method. Some mathematical formulations covered in the original paper will not be included for simplicity. Illustration of the graphic model is shown in Figure 1.

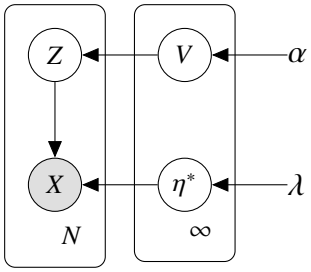


Figure 1: Graphical Model

In the method section of the original paper, $p(X|\eta^*)$ and $p(\eta^*|\lambda)$ are mentioned as exponential family distributions without giving specification. Here for better illustration, we consider them as Gaussian-Gaussian distributions (with known and fixed covariance matrix Σ) based on the form of

data in the experiments. We can transform the formulation of Gaussian distribution to exponential family representation as:

$$\begin{aligned} f_\Sigma(x|\mu) &= (2\pi)^{-\frac{k}{2}}\det(\Sigma)^{-\frac{1}{2}}\exp\{-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\} \\ &= (2\pi)^{-\frac{k}{2}}\det(\Sigma)^{-\frac{1}{2}}\exp\{-\frac{1}{2}\left(x^T\Sigma^{-1}x - 2\mu^T\Sigma^{-1}x + \mu^T\Sigma^{-1}\mu\right)\} \\ &= (2\pi)^{-\frac{k}{2}}\det(\Sigma)^{-\frac{1}{2}}\exp\{-\frac{1}{2}x^T\Sigma^{-1}x\}\cdot\exp\{\mu^T\Sigma^{-1}x - \frac{1}{2}\mu^T\Sigma^{-1}\mu\} \\ &= h(x)\exp\{\ \eta^{*T}x - a(\eta^*)\} \end{aligned}$$

where we denote:

$$\begin{aligned} h(x) &= (2\pi)^{-\frac{k}{2}}\det(\Sigma)^{-\frac{1}{2}}\exp\{-\frac{1}{2}x^T\Sigma^{-1}x\} \\ \eta^* &= (\mu^T\Sigma^{-1})^T = \Sigma^{-1}\mu \\ a(\eta^*) &= \frac{1}{2}\eta^{*T}\Sigma\eta^* \end{aligned}$$

The conjugate prior follows the form of Gaussian $\mathcal{N}(\mu_0, \Sigma_0)$ and can also be expressed as exponential family representation as:

$$\begin{aligned} p(\eta^*|\lambda) &= (2\pi)^{-\frac{k}{2}}\det(\Sigma_0)^{-\frac{1}{2}}\exp\{-\frac{1}{2}\eta^{*T}\Sigma_0^{-1}\eta^* + \mu_0^T\Sigma_0^{-1}\eta^* - \frac{1}{2}\mu_0^T\Sigma_0^{-1}\mu_0\} \\ &= (2\pi)^{-\frac{k}{2}}\det((\lambda_2\Sigma)^{-1})^{-\frac{1}{2}}\exp\{-\frac{1}{2}\eta^{*T}(\lambda_2\Sigma)\eta^* + \lambda_1^T\eta^* - \frac{1}{2}\mu_0^T\Sigma_0^{-1}\mu_0\} \\ &= h(\eta^*)\exp\{\lambda_1^T\eta^* + \lambda_2(-a(\eta^*)) - a(\lambda)\} \end{aligned}$$

where we denote $\Sigma_0 = \Sigma^{-1}/\lambda_2,\ \mu_0 = \Sigma_0\lambda_1$ and

$$\begin{aligned} h(\eta^*) &= (2\pi)^{-\frac{k}{2}}\det(\Sigma)^{\frac{1}{2}} \\ a(\lambda) &= \frac{1}{2}\lambda_1^T(\Sigma^{-1}/\lambda_2)\lambda_1 - \frac{D}{2}\log\lambda_2 \end{aligned}$$

Note the derivation results are in align with the form mentioned in Blei and Jordan [2], a subsequent paper of the authors.

With the base distributions set, we move on to the part of variational inference which, in the paper, the factorized variational distribution is represented as:

$$q(v, \eta^*, z, K) = \prod_{i=1}^{K-1} q(v_i|\gamma_i)\prod_{i=1}^{K} q(\eta_i^*|\tau_i)\prod_{n=1}^{N} q(z_n|\phi_n)$$

And due to the property of conjugate prior, the factorized distributions have the same form as the original which are formulated as:

$$\begin{aligned} q(v_i|\gamma_i) &= Beta(\gamma_{i,1}, \gamma_{i,2}) \\ q(\eta_i^*|\tau_i) &= h(\eta_i^*)\exp\{\tau_{i,1}^T\eta_i^* + \tau_{i,2}(-a(\eta_i^*)) - a(\tau_i)\} = \mathcal{N}(\mu', \Sigma') \\ q(z_n = i) &= \phi_{n,i} \end{aligned}$$

The optimization is based on coordinate ascent algorithm that iterative updates the parameters between ϕ and others (e.g. γ, τ). We first show the update functions of the latter which is:

$$\gamma_{i,1} = 1 + \sum_n \phi_{n,i} \qquad \gamma_{i,2} = \alpha + \sum_n \sum_{j=i+1}^{K} \phi_{n,j}$$
$$\tau_{i,1} = \lambda_1 + \sum_n \phi_{n,i} x_n \qquad \tau_{i,2} = \lambda_2 + \sum_n \phi_{n,i}$$

For ϕ we have $\phi_{n,i} \propto \exp(E)$, where:

$$E = \mathbb{E}[\log V_i|\gamma_i] + \mathbb{E}[\eta_i|\tau_i]^T X_n - \mathbb{E}[a(\eta_i)|\tau_i] + \sum_{j=1}^{i-1} \mathbb{E}[\log(1 - V_j)|\gamma_j]$$

For $\mathbb{E}[\log V_i|\gamma_i]$ and $\mathbb{E}[\log(1 - V_i)|\gamma_i]$ we have:

$$\mathbb{E}[\log V_i|\gamma_i] = \Psi(\gamma_{i,1}) - \Psi(\gamma_{i,1} + \gamma_{i,2})$$
$$\mathbb{E}[\log(1 - V_i)|\gamma_i] = \Psi(\gamma_{i,2}) - \Psi(\gamma_{i,1} + \gamma_{i,2})$$

where $\Psi(\cdot)$ is the digamma function. For $\mathbb{E}[\eta_i|\tau_i]$, we have:

$$\mathbb{E}[\eta_i|\tau_i] = \mu' = (\tau_{i,2}\Sigma)^{-1}\tau_{i,1}$$

For $\mathbb{E}[a(\eta_i)|\tau_i]$, we have:

$$\begin{aligned}
\mathbb{E}[a(\eta_i)|\tau_i] &= \mathbb{E}[\frac{1}{2}\eta^{*T}\Sigma\eta^*|\tau_i] = \mathbb{E}[tr(\frac{1}{2}\eta^{*T}\Sigma\eta^*)|\tau_i] \\
&= \mathbb{E}[tr(\frac{1}{2}\Sigma\eta^*\eta^{*T})|\tau_i] = \frac{1}{2}tr(\Sigma\mathbb{E}[\eta^*\eta^{*T}|\tau_i]) \\
&= \frac{1}{2}tr(\Sigma(\Sigma' + \mu'\mu'^T)) = \frac{1}{2}tr(\Sigma(\tau_{i,2}\Sigma)^{-1}) + \frac{1}{2}\mu'^T\Sigma\mu' \\
&= \frac{1}{2\tau_{i,2}}tr(\Sigma^{-1}\Sigma) + \frac{1}{2}\mu'^T\Sigma\mu' = \frac{D}{2\tau_{i,2}} + \frac{1}{2}\mu'^T\Sigma\mu'
\end{aligned}$$

where D is the dimension of Σ, in which case also the data dimension. And up till now, we have shown the update steps in the VI model. We first update γ and τ with ϕ, and then iteratively update ϕ with γ and τ.

Held-out Log-Likelihood

For model inference, we show the calculation process of the predictive distribution or the held-out log-likelihood. That is to predict the probability of a new data point given the trained parameters:

$$p(x_{N+1}|\mathbf{z}, \alpha, \lambda) = \sum_{i=1}^{K} \mathbb{E}[\theta_i|\gamma] p(x_{N+1}|\tau_i)$$

For $p(x_{N+1}|\tau_i)$ we have:

$$
\begin{aligned}
p(x_{N+1}|\tau_i) &= \int_{\eta_i} p(x_{N+1}, \eta_i|\tau_i) = \int_{\eta_i} p(x_{N+1}|\eta_i)p(\eta_i|\tau_i) \\
&= \int_{\eta_i} h(x_{N+1})\exp\{\eta_i^T x_{N+1} - a(\eta_i)\} \cdot h(\eta_i)\exp\{\tau_{i,1}^T \eta_i + \tau_{i,2}(-a(\eta_i)) - a(\tau_i)\} \\
&= \int_{\eta_i} h(x_{N+1})h(\eta_i)\exp\{(x_{N+1} + \tau_{i,1})^T \eta_i - (\tau_{i,2}+1)a(\eta_i) - a(\tau_i)\} \\
&= h(x_{N+1}) \frac{\exp\{\mathrm{a}(\begin{pmatrix} x_{N+1} + \tau_{i,1} \\ 1 + \tau_{i,2} \end{pmatrix})\}}{\exp\{\mathrm{a}(\begin{pmatrix} \tau_{i,1} \\ \tau_{i,2} \end{pmatrix})\}}
\end{aligned}
$$

For $\mathbb{E}[\theta_i|\gamma]$, we have:

$$
\begin{aligned}
\mathbb{E}[\theta_i|\gamma] &= \mathbb{E}[V_i \prod_{j=1}^{i-1}(1 - V_i)|\gamma] \\
&= \mathbb{E}[V_i|\gamma] \prod_{j=1}^{i-1} \mathbb{E}[1 - V_i|\gamma] \\
&= \frac{\gamma_{i,1}}{\gamma_{i,1} + \gamma_{i,2}} \prod_{j=1}^{i-1} \frac{\gamma_{j,2}}{\gamma_{j,1} + \gamma_{j,2}}
\end{aligned}
$$

Results

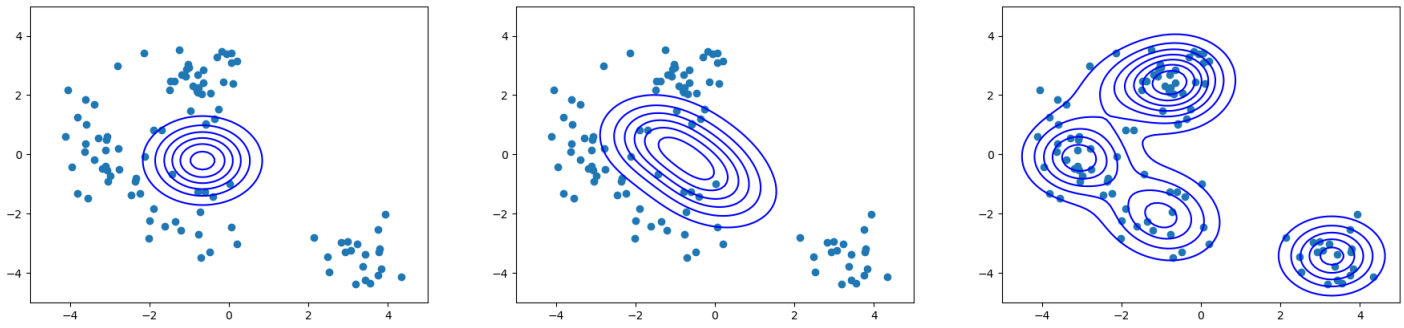
Toy example

Following the paper, we illustrate the effectiveness of the VI algorithm through a toy problem setting where data is drawn from 2-Dimensional Gaussian distributions with diagonal covariance. Through each iteration, the predictive distribution is shown in Figure 2. Since we don't know the exact initialization method of the paper, we simply set all parameters to 1 for initialization. This could cause the algorithm to iterate more for convergence. But in the end, the model still converges well enough to fit the data and we report the same thing that the model only uses 5 components even though we set K to 20.

Simulation experiments

As mentioned in the paper, we generate 200 data points from a Gaussian-Gaussian DP mixture model, where the first 100 points are used as $\mathbf{X}$ for training VI algorithms and the last 100 points are treated as held-out data. The fixed covariance matrix Σ of the data is given by

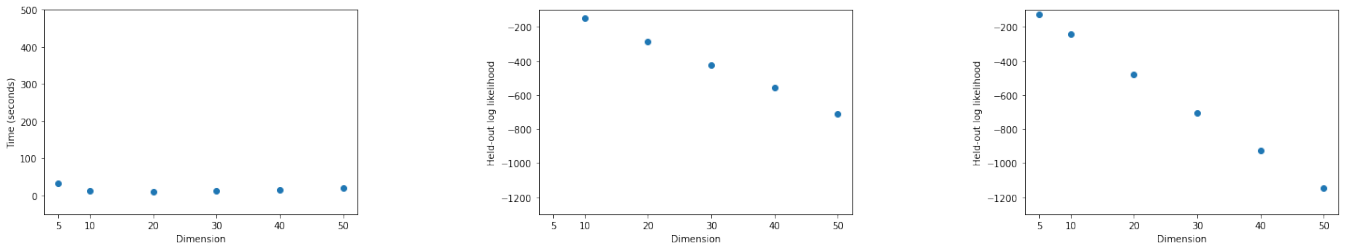
$$
\Sigma_{i,i} = 1,\ \Sigma_{i,i+1} = \Sigma_{i,i-1} = c,\ i = 1, ..., d; \Sigma_{i,j} = 0\ if\ |i - j| \geq 2
$$



(a) Iteration 0 (b) Iteration 5 (c) Iteration 50

Figure 2: The toy example of VI

The baseline distribution $\eta^* \sim \mathcal{N}(\Sigma^{-1}\lambda_1/\lambda_2, \Sigma^{-1}/\lambda_2)$, Then $\mu = \Sigma\eta^* \sim \mathcal{N}(\lambda_1/\lambda_2, \Sigma/\lambda_2)$. The number of clusters/truncation level is K. In our experiments, we set $c = 0.4, K = 20, \lambda_1 = \mathbf{0}, \lambda_2 = 1$, and conduct experiments in terms of different data dimensions. For each kind of dimension, we choose α from the list $[0.1, 0.2, 0.3, 0.5, 0.8, 1, 5, 10, 100, 1000]$ to form 10 different datasets, which is the hyperparameter for variable V_i when constructing a DP process.



(a) Total running time(10 dataset) (b) Original setting (c) New setting

Figure 3: Results of simulated experiments

Running Time As we can see in Figure 3a, the VI algorithm is really fast, and the result is similar to that in the paper, the increasing dimension does not slow down its speed.

Held-out Log likelihood For each kind of dimension, first, we calculate the average log-likelihood for each dataset and then sum them up. At first, we follow the settings mentioned above and obtain the result in Figure 3b. In general, the tendency that as the dimension increases log-likelihood decreases is the same. The only difference is that our results here are a little bit greater. We think this is because many settings are not explicit in the paper, such as the choice for c, α, λ_2, etc. Then we try tunning some hyperparameters, and finally through multiplying the covariance matrix Σ by 10 with all other parameters unchanged, we obtain the new results in Figure 3, which is very close to that in the paper. So for the simulation part, we conclude the results are reproducible.

Robot data experiments

We also collect real-world data from the Pumadyn robot dataset proposed by Ueda and Ghahramani [3]. Following the paper, we split the data points into 7000 training samples and keep 250 test samples for the predictive distribution test. The mean and covariance in the hyperparameters are set as the actual mean and covariance calculated from the training samples. The final results are displayed in Table 1. The held-out likelihood is noticeably larger than that shown in the paper. We hypothesize that this is related to the initialization, hyperparameter settings, and even the preprocessing steps for the datasets since none of the above is specified in the paper. In this case, the robot experiment is only weakly reproducible. Interestingly, it takes significantly less time for training thanks to the development of modern computers.

Table 1: Results on robot data

Time(Sec.)	Held-out like.	# Comp.
10.93	−2420.09	7

Discussion

In the paper, we showed the derivations of variational inference on Gaussian-Gaussian DP mixture models and analyzed the method on a toy example, simulated data, and real-world robot data. The method is proposed to accelerate the training and inference phase by adopting deterministic approximations toward the posterior and the experiments have indeed proven the performance boost. However, the parameterized approach also induces certain drawbacks such as easily reached local maxima – initialization of the parameters must be carefully taken care of especially in the real-world data that is more chaotic and with more noise compared to the synthetic ones. And the method is restricted to the data and prior distributions – so far, only exponential family distributions are introduced for the algorithm which can be limited in usage. The fact that we still need to specify K indicates that further improvement can be done. Nonetheless, the method is new and provides applicable alternatives to the MCMC methods.

References

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[3] Naonori Ueda and Zoubin Ghahramani. Bayesian model search for mixture models based on optimizing variational bounds. *Neural Networks* 15.10 (2002), 1223–1241.